Syllabus: DX603 Module 3
Machine Learning Fundamentals
3 credits

Module 3 Dates

Mod 3 starts on January 16 and runs through May 2, 2026.

Module 3 Faculty

Professor Wayne Snyder

In addition, Module 3 content and expertise contributed by Giorgos Zervas and Jef Considine.

Module 3 Learning Facilitator

Vishnupriya Kasireddy

Your assigned Learning Facilitator is your main contact during Module 3 for course-related matters pertaining to all course assignments, grades, and concerns directly arising within the module (i.e., course materials and supplies). To communicate with your Learning Facilitator, go to BU Virtual Campus.

Throughout Module 3, Learning Facilitators will hold **group office hours**, and students may attend as many group office hours as they wish. Learning Facilitators are also available for individual appointments. An appointment can be requested by submitting a Help Request via the BU Virtual Campus. **Please be mindful that there are a limited number of available slots for individual appointments. Therefore, use this option judiciously.**

Student Success Specialists

Annie Imperato

Your Student Success Specialist will support your progression through Module 3 and the OMDS program in general by providing valuable resources to aid your learning journey. Your Success team functions similarly to an academic coach or advisor, and

this team will help you get registered and support extracurricular programming. You may reach the Student Success team at BU Virtual Campus.

Module 3 Overview and Learning Objectives

Module 3 covers the Fundamentals of Machine Learning and best practices for using regression, tree, and ensemble methods for classification and regression problems. It is part of a two-module sequence that leads to Deep Learning in Module 7.

During Module 3, you will learn to:

1. **Understand the Core Concepts of Machine Learning**
   Explain the key principles of machine learning, including its taxonomy (supervised, unsupervised, and reinforcement learning) and the distinction between prediction and inference.

2. **Apply and Evaluate Machine Learning Algorithms**
   Develop and implement machine learning algorithms to address problems involving regression, classification, clustering, and dimensionality reduction, gaining experience with specific models such as linear regression, multiple regression, polynomial regression, lasso regression, ridge regression, logistic regression, decision trees, and ensemble methods like Random Forests and Boosting.

3. **Analyze and Prepare Data for Machine Learning Models**
   Demonstrate skills in data cleaning, feature engineering, and exploratory data analysis to ensure data quality and suitability for machine learning applications.

4. **Assess Model Performance and Generalization**
   Use metrics and validation techniques such as cross-validation and train-test splits to evaluate and improve model performance and generalization.

5. **Address Overfitting and Underfitting**
   Explain and mitigate issues of overfitting and underfitting through methods such as model selection, regularization, and pruning.

6. **Navigate Trade-Offs in Model Design**
   Balance competing objectives such as accuracy, interpretability, and fairness in the design and deployment of machine learning models.

7. **Recognize and Mitigate Bias and Fairness Issues**
   Identify sources of bias in data and models, and apply strategies to improve fairness, including fairness metrics and bias mitigation techniques.

8. **Communicate Results Effectively**
   Interpret and present machine learning results in a clear, concise, and meaningful way, tailored to technical and non-technical audiences.

Module 3 Weekly Schedule

Release Dates	Concepts	Live Sessions
Week 1: January 16	Introduction to Machine Learning	Tuesday January 20 6pm ET **(Tuesday session due to MLK Day)**
Week 2: January 23	Linear Regression	Monday January 26, 6pm ET
Week 3: January 30	Generalization, Errors, and Bias-Variance Tradeoff	Monday February 2, 6pm ET
Week 4: February 6	Generalization; Sampling Methods: Training, Testing, and Validation Sets; Cross-Validation	Monday February 9, 6pm ET
Week 5: February 13	Feature Selection and Regularization	Tuesday February 17, 6pm ET **(Tuesday session due to Presidents' Day)**
Week 6: February 20	Decision Trees	Monday February 23, 6pm ET
Week 7: February 27	Ensemble Methods: Bagging, Boosting, and Random Forests	Monday March 2, 6pm ET

Spring Recess	**March 7 - 15**	
Week 8: March 6	Classification	Monday March 16, 6pm ET **(delayed due to recess)**
Week 9: March 20	Variable Importance Measures	Monday March 23, 6pm ET
Week 10: March 27	Causal Inference	Monday March 30, 6pm ET
Week 11: April 3	Bias and Fairness	Monday April 6, 6pm ET
Week 12: April 10	Final Project Management	Monday April 13, 6pm ET
Week 13: April 17	Unsupervised Learning	Tuesday April 21, 6pm ET **(Tuesday session due to Patriots' Day)**
Week 14: April 24	Deep Learning	Monday April 27, 6pm ET

Module 3 Live Sessions

Live Sessions for Module 3 are offered each Monday during the term. Because Live Sessions are an integral part of the program and are offered weekly, you are strongly encouraged to attend all Live Sessions.

All sessions are offered at 6pm ET unless otherwise noted. If you are unable to attend a Live Session, please watch the recording. Professors consider Live Sessions

essential learning, and content covered in the Live Session is eligible to be included in Module assessments.

The teaching team requests that you keep your video/web cameras on during Live Sessions (unless otherwise noted, or you may turn off your camera due to bandwidth problems).

Community Standards for Live Sessions

Live Sessions in the OMDS program are an active learning environment and include actual, real-time interactions (teaching team to student, student to student). Adherence to the Student Codes of Conduct is expected. Additionally, the program recognizes that open discourse and authentic representation of the lived experience are valuable in the learning environment. Ideas and ideologies can be framed respectfully while we all seek first to understand as well as be understood. Because OMDS encourages students to be authentic, this approach suggests that honest mistakes are possible. When this happens, learning from each other is essential. As a result, OMDS strives not to censor language or discourse during the Live Session but to demonstrate the learning opportunity as needed.

Homework Assignments

The first eight weeks of the module consist of eight weekly assignments which exercise your understanding of that week's content and develop competency in building and evaluating machine learning models.

Final Project

Module 3 Final Project will apply the techniques learned throughout the term to a suitable dataset. Teams of three will be randomly chosen, and you will have a selection of datasets and problems to choose from.

Students will work through three phases:

1. **Milestone 1:** Define the problem, identify data sources, and perform exploratory data analysis (EDA) to understand the dataset and its challenges.

2. **Milestone 2:** Develop and evaluate machine learning models, including a baseline and at least two advanced models, while refining features and hyperparameters.
3. **Final Report:** Present a comprehensive analysis of the problem, data insights, modeling process, evaluation results, and practical implications. Teams will complete a peer review to assess individual contributions.

This project emphasizes critical thinking, teamwork, and practical application of machine learning concepts.

There is a total of 500 points in Module 3.

Grading

Throughout Module 3, you will complete activities for credit in the various gradebook assessment categories listed below. Please make sure that you complete all your activities accordingly.

Calculation of final homework average: There are eight weekly homework assignments, which are an essential part of your learning for the module. However, to accommodate occasional scheduling conflicts, we will drop the lowest homework score. No part of the final project may be skipped.

We have broken the various gradebook assessment categories out by academic content, key competencies, and peer engagement. Here's how each category contributes to your overall grade:

Category	**Grading Breakdown**
Academic Content	500 points = 100%
• 7 Weekly Assignments @ 55 pts each	7 * 55 = 385 = 77%
• Final Project	115 pts = 23%
• Milestone One	25 pts = 5%
• Milestone Two	40 pts = 8%
• Final submission	50 pts = 10%

Policy Regarding Late Submissions

The OMDS program is committed to student success and strives to maintain the highest level of performance and integrity of the degree. All OMDS assignments (solo or team) are due at 11:59pm ET on Sunday night, with a two-hour grace period.

Otherwise, the following late policy applies:

- All assignments are due at 11:59pm ET for full credit, with a grace period of two hours. The grace period only applies to the original deadline and does not apply to subsequent partial credit deadlines.
- Assignments submitted more than two hours and up to five days late will get partial credit but be penalized 10% per day for lateness.
- Assignments submitted more than five days (120 hours) late will receive zero points.
- Specifically:
    - Within zero to two hours of original deadline: no penalty
    - Within two to 24 hours of original deadline: 90% of grade
    - Within 24-48 hours of original deadline: 80% of grade
    - Within 48-72 hours of original deadline: 70% of grade
    - Within 72-96 hours of original deadline: 60% of grade
    - Within 96-120 hours of original deadline: 50% of grade
    - More than 120 hours late: 0% of grade

Regarding incompletes, we follow BU's incomplete policy

In the case of an emergency, students should submit a case via BU Virtual Campus to their assigned Learning Facilitator as soon as possible providing any relevant documentation. The Learning Facilitator will work with the Student Success team and faculty to determine if an exception to the late submission policy can be made.

Course Materials

- All textbooks and course materials are free, either online or through the BU Library

- Students are encouraged to join Medium (blogs about machine learning, $5/month) or another paid subscription service and use chatGPT Plus for research; the cost is $20/month

Technology Advice for OMDS

For the best technical experience, students are encouraged to use a laptop or desktop computer running Mac OS X or Windows. Recommended browsers include Firefox or Chrome – access this link for more information on browser compatibility and other system requirements. In the event of technical difficulties, you can try an alternate browser or submit a help desk ticket to request help troubleshooting the problem.

While course materials are accessible from mobile devices, not all tools are optimized for these devices , and we would strongly recommend using a laptop or desktop to complete assignments and other key components of your coursework.

Requesting Accommodations

In accordance with university policy, every effort will be made to accommodate unique and special needs of students with respect to speech, hearing, vision, or other disabilities. Any student who feels they may need accommodation for a documented disability should contact BU Disability & Access Services online or by phone at 617-353-3658 for review and approval of accommodation requests. It is best to start this process as early as possible.

Sexual Misconduct/Title IX Policy

Boston University is committed to fostering a safe learning environment for all members of its community and preventing sexual misconduct. All forms of sexual misconduct, including rape, acquaintance rape, sexual assault, domestic and dating violence, stalking, and sexual harassment, are violations of Boston University's policies, whether they happen on campus or off campus.

Title IX of the Education Amendments Act of 1972 is a federal civil rights law that prohibits sex- based discrimination in federally funded education programs and

activities. This law makes it clear that violence and harassment based on sex and gender are civil rights offenses subject to the same kinds of accountability and the same kinds of support applied to offenses against other protected categories such as race, national origin, etc. If you or someone you know has been harassed or assaulted, you can find the appropriate resources at Sexual Misconduct Safety.

Academic Integrity

Students are expected to do their own work, not cheat, and must abide by Boston University's Academic Conduct Code

> *"Boston University is a community of learning grounded in shared values, rights, and responsibilities. One critical component of our learning community is academic integrity, that is a commitment, even in the face of adversity, to the fundamental values of honesty, trust, fairness, respect, responsibility, and courage in representing truthfully one's knowledge and ideas as one's own. Our community can only be maintained if all students represent honestly their own knowledge and ideas, believe that their academic competence is being judged fairly based on their knowledge and ideas, and have confidence that they will not be put at a disadvantage because of someone else's dishonesty. Promoting and protecting academic integrity is the responsibility of every member of the campus community because our goals of teaching, learning, and research can only be accomplished in environments in which ethical standards are upheld.*
>
> *"Academic dishonesty undermines the bonds of trust and honesty between members of the community and defrauds those who depend upon our knowledge and integrity. Students, faculty, and staff of Boston University support this Academic Conduct Code for the purposes of providing an atmosphere of mutual trust and respect, fostering ethical behavior, and cultivating lifelong professional conduct."*

Solo assignments must be completed with solo effort, and any and all work must have proper citation for all OMDS assignments. Team assignments may only be shared with your designated team. The OMDS program utilizes Turnitin (a plagiarism detection software) to assess submitted assignments. Any student believed to be in violation of the Academic Conduct Code for any reason will be subject to the university process as outlined at Academic Conduct Code.

Students should learn how to use AI text generators and other AI-based assistive resources (collectively, AI tools) to enhance rather than damage their developing abilities as writers, coders, communicators, and thinkers. The CDS Generative AI

Assistance (GAIA) Policy provides guidance on how to use AI tools safely and responsibly while adhering to the Academic Conduct Code.

Community Code of Conduct

As members of the OMDS community, each of us has a collective and individual responsibility to conduct ourselves in accordance with Boston University's core values:

1. Collaborate with each other and the teaching team (i.e., faculty and Learning Facilitator team) to foster a culture of inclusion.
2. Treat each other with dignity, fairness, and respect.
3. Communicate using a professional tone and clarity to model the behavior of a business leader.
4. Debate respectfully in online forums.
5. Give feedback directly to others respectfully and constructively.

Violations of the community code of conduct pose a challenge to the overall student code of conduct and may be subject to administrative response and consequence. Students are valued members of the OMDS community and can be assured that OMDS faculty and staff also adhere to the same community code of conduct.

*The OMDS program reserves the right to modify the requirements of the Mod with adequate notice to students and within the scope of authority and responsibility entrusted by Boston University.